

Laplace Transform

Script: Chapter 7 (Laplace transform, transfer functions, stability)

with solutions

Learning goals

- Describe engine sensors by transfer functions $H(s)$; relate poles and zeros to decay time, resonance frequency, damping, Q and bandwidth.
- Compute impulse and step responses by partial fractions and decide stability from the pole locations.
- Read and sketch Bode diagrams; judge an RC anti-aliasing filter.
- Handle a transport delay e^{-sT_d} by Padé approximation and understand the consequences of a right-half-plane zero.
- Verify analytic results by numerical ODE integration (Euler, RK4) and understand the stability limit of explicit integrators.
- Map an analog system into the z -domain (forward Euler, impulse invariance, bilinear transform) – a preview of Sheets 9 and 10.

Background

For a causal signal $x(t)$ the (unilateral) Laplace transform is $X(s) = \int_0^\infty x(t) e^{-st} dt$. An LTI system described by a linear ODE with constant coefficients has the *transfer function*

$$H(s) = \frac{Y(s)}{X(s)} = \frac{b_0 s^m + \dots + b_m}{a_0 s^n + \dots + a_n} = K \frac{\prod_k (s - z_k)}{\prod_k (s - p_k)},$$

the Laplace transform of its impulse response $h(t)$. A causal system is BIBO stable iff all poles satisfy $\operatorname{Re} p_k < 0$. On the imaginary axis, $s = j\omega$, $H(j\omega)$ is the frequency response.

The knock burst of the virtual engine (`code/common/engine_signals.h`) is

$$x_k(t) = A e^{-t/\tau} \sin(2\pi f_d t) \varepsilon(t), \quad f_d = 6.5 \text{ kHz}, \quad \tau = 0.8 \text{ ms}$$

(plus a second mode at 10.5 kHz). We interpret it as the impulse response of the mechanical path *combustion chamber* $\rightarrow$ *engine block* $\rightarrow$ *piezo sensor*, which is excited by the short pressure shock of knocking combustion. Throughout this sheet we use the normalised second-order model ($\sigma = 1/\tau$, $\omega_d = 2\pi f_d$)

$$H_K(s) = \frac{\omega_0^2}{s^2 + 2\sigma s + \omega_0^2}, \quad \omega_0^2 = \omega_d^2 + \sigma^2, \quad \zeta = \sigma/\omega_0. \quad (1)$$

Exercise 7.1 Sensor models in the s -domain

PAPER

- Knock resonator.** Show that the impulse response of (1) is $h_K(t) = \frac{\omega_0^2}{\omega_d} e^{-\sigma t} \sin(\omega_d t) \varepsilon(t)$. Compute σ , ω_0 (and f_0), ζ , the quality factor $Q = 1/(2\zeta)$, the poles and the region of convergence. Is the system stable? What would $\sigma < 0$ mean?
- Compute the **step response** of H_K with partial fractions. Give the overshoot, the time of the first maximum and the 2% settling time. The DC gain is $H_K(0) = 1$: what does this imply for the slow combustion vibration (a half sine over 60° crank angle) that is also present on the sensor signal?
- Bode diagram.** Sketch $|H_K(j\omega)|$ in dB and the phase over $\log f$ (asymptotes!). Compute the height and frequency of the resonance peak and show that the -3 dB bandwidth is $\Delta\omega \approx 2\sigma$, i.e. $B \approx 1/(\pi\tau)$ in Hz. Interpret this relation between decay time and bandwidth.

- d) The second knock mode has $f_d = 10.5$ kHz and the same τ . Mark all four poles of the two-mode model in the s -plane. What distinguishes the poles of the two modes, and what do they have in common?
- e) **RC anti-aliasing filter.** The knock signal is sampled at $f_s = 40$ kHz behind an RC low-pass with $R = 1$ k Ω , $C = 10$ nF. Give $H_{RC}(s)$, its pole and f_c . Compute the attenuation at 6.5 kHz, 10.5 kHz and at 30 kHz. Onto which frequency does a 30 kHz component alias, and why is that critical? Which f_c would a first-order filter need for 40 dB attenuation at 30 kHz? Compare with a second-order Butterworth filter with the same f_c and conclude. Also give the phase and the group delay of the RC filter at 6.5 kHz.
- f) **Lambda sensor with transport delay.** Fuel change $\rightarrow$ sensor is modelled as $G(s) = e^{-sT_d}/(1 + sT)$ with $T_d = 200$ ms (gas travel) and $T = 30$ ms (sensor), the values of the Wokwi lambda-probe chip.
- Why is $G(s)$ not a rational function, and why is this a problem for the methods of this chapter?
 - Derive the Padé(1,1) approximation $e^{-sT_d} \approx (1 - sT_d/2)/(1 + sT_d/2)$ by matching the Taylor series of both sides as far as possible. Where are poles and zeros of the approximated $G_1(s)$?
 - Compute the step response of $G_1(s)$ with partial fractions. Determine the minimum (time and value) and explain the *undershoot*. Compare with the exact step response.
 - Phase error: the exact delay has phase $-\omega T_d$, the Padé term $-2 \arctan(\omega T_d/2)$. Up to which frequency is the error below 5°? (Use the series $\arctan x \approx x - x^3/3$.)

Solution

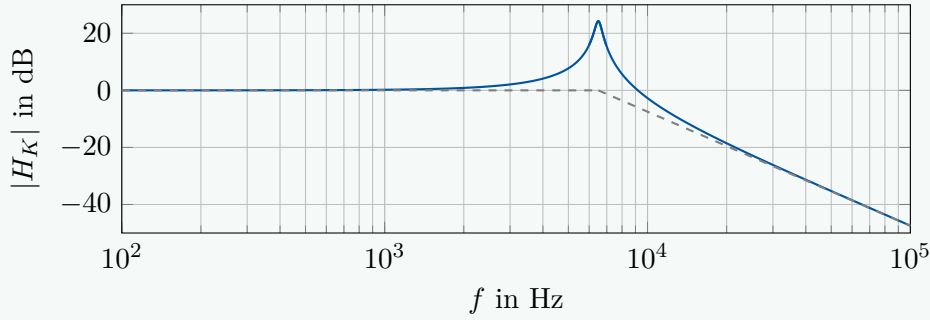
a) With $\mathcal{L}\{e^{-\sigma t} \sin(\omega_d t) \varepsilon(t)\} = \omega_d / ((s + \sigma)^2 + \omega_d^2)$ and $(s + \sigma)^2 + \omega_d^2 = s^2 + 2\sigma s + \sigma^2 + \omega_d^2 = s^2 + 2\sigma s + \omega_0^2$ we get $\mathcal{L}\{h_K\} = \frac{\omega_0^2}{\omega_d} \cdot \frac{\omega_d}{s^2 + 2\sigma s + \omega_0^2} = H_K(s)$. Numbers: $\sigma = 1/\tau = 1250$ s $^{-1}$, $\omega_d = 40\,840.7$ rad/s, $\omega_0 = 40\,859.8$ rad/s ($f_0 = 6503.0$ Hz), $\zeta = 0.0306$, $Q = 16.3$. Poles $p_{1,2} = -\sigma \pm j\omega_d = -1250 \pm j40841$ s $^{-1}$, ROC $\text{Re } s > -1250$ s $^{-1}$. It contains the imaginary axis: the causal system is stable. $\sigma < 0$ means poles in the right half plane: the oscillation grows like $e^{|\sigma|t}$ (unstable).

b) $\frac{H_K(s)}{s} = \frac{1}{s} - \frac{s+2\sigma}{(s+\sigma)^2 + \omega_d^2} = \frac{1}{s} - \frac{s+\sigma}{(s+\sigma)^2 + \omega_d^2} - \frac{\sigma}{\omega_d} \frac{\omega_d}{(s+\sigma)^2 + \omega_d^2}$, hence

$$y_\varepsilon(t) = \left[1 - e^{-\sigma t} \left(\cos \omega_d t + \frac{\sigma}{\omega_d} \sin \omega_d t \right) \right] \varepsilon(t).$$

First maximum at $t_p = \pi/\omega_d = 76.9$ μ s, overshoot $e^{-\pi\sigma/\omega_d} = e^{-0.0962} = 0.908$, i.e. **90.8 %**; 2 % settling time $\approx 4/\sigma = 3.2$ ms. Because $H_K(0) = 1$ the slow combustion vibration (half sine of 3.3 ms at 3000 rpm, i.e. some 100 Hz) passes unattenuated – the resonator alone is no knock filter; a band-pass (Sheet 10) or at least a high-pass is required.

c) Asymptotes: 0 dB below f_0 , -40 dB/decade above f_0 (two poles); phase $0^\circ \rightarrow -180^\circ$, -90° at f_0 , the transition takes place within $\approx \pm B/2$ around f_0 . Peak: $|H|_{\max} = 1/(2\zeta\sqrt{1-\zeta^2}) = 16.35$ (24.27 dB) at $f_r = f_0\sqrt{1-2\zeta^2} = 6497$ Hz. Near the pole p_1 , $|H| \propto 1/|j\omega - p_1| = 1/\sqrt{\sigma^2 + (\omega - \omega_d)^2}$, which drops by $\sqrt{2}$ for $|\omega - \omega_d| = \sigma$: $\Delta\omega = 2\sigma$, $B = \sigma/\pi = 1/(\pi\tau) = 398$ Hz. A short burst (small τ) necessarily has a wide spectrum and vice versa (time–bandwidth product $B\tau = 1/\pi$). Numerical check in Exercise 7.2: $B = 398.3$ Hz.



d) Mode 2: $p_{3,4} = -1250 \pm j65973 \text{ s}^{-1}$. Same real part (same decay time τ , i.e. same distance from the imaginary axis, same bandwidth 398 Hz), but larger imaginary part (frequency); its damping ratio $\zeta_2 = \sigma/\omega_{0,2} = 0.019$ and $Q_2 = 26$ are therefore different.

e) $H_{RC}(s) = 1/(1 + sRC)$, $RC = 10 \mu\text{s}$, pole $s = -10^5 \text{ s}^{-1}$, $f_c = 1/(2\pi RC) = 15.9 \text{ kHz}$. $|H|$: -0.67 dB at 6.5 kHz , -1.57 dB at 10.5 kHz , -6.58 dB at 30 kHz . At $f_s = 40 \text{ kHz}$ a 30 kHz component aliases to $40 - 30 = 10 \text{ kHz}$, right next to the second knock mode, and is attenuated by a factor of only 0.47. For 40 dB : $\sqrt{1 + (30/f_c)^2} = 100 \Rightarrow f_c \approx 300 \text{ Hz}$ – that would also destroy the knock band. A second-order Butterworth with $f_c = 15.9 \text{ kHz}$ gives $-10 \log_{10}(1 + (f/f_c)^4)$: -0.12 dB at 6.5 kHz and -11.3 dB at 30 kHz ; still not enough. Conclusion: either a higher-order analog filter or (better) oversampling with a digital decimation filter. Phase at 6.5 kHz : $-\arctan(\omega RC) = -22.2^\circ$; group delay $\tau_g = RC/(1 + (\omega RC)^2) = 8.6 \mu\text{s}$ ($\approx 0.15^\circ$ crank angle at 3000 rpm – negligible).

f) i. $\mathcal{L}\{x(t - T_d)\} = e^{-sT_d}X(s)$ is a transcendental function of s ; it cannot be written as a ratio of polynomials, so there are no finite sets of poles and zeros, no partial fraction expansion and no finite-dimensional ODE / state-space model (the “state” of a delay is the complete input history of length T_d).

ii. $e^{-x} = 1 - x + \frac{x^2}{2} - \frac{x^3}{6} + \dots$ and $\frac{1-ax}{1+ax} = 1 - 2ax + 2a^2x^2 - 2a^3x^3 + \dots$; matching the linear and the quadratic term gives $a = \frac{1}{2}$ (the error starts with $x^3/12$). $G_1(s) = \frac{1-0.1s}{(1+0.1s)(1+0.03s)}$: poles $s = -10$ and $s = -33.3 \text{ s}^{-1}$, zero $s = +10 \text{ s}^{-1}$ in the right half plane (non-minimum phase, all-pass factor).

iii. $\frac{G_1(s)}{s} = \frac{1-0.1s}{0.003s(s+10)(s+100/3)} = \frac{1}{s} - \frac{20/7}{s+10} + \frac{13/7}{s+100/3}$:

$$y(t) = 1 - \frac{20}{7}e^{-10t} + \frac{13}{7}e^{-100t/3} \quad (t \text{ in s}).$$

$y(0) = 0$, $\dot{y}(0) = \frac{200}{7} - \frac{1300}{21} = -33.3 \text{ s}^{-1} < 0$: the response starts in the *wrong direction* (typical of an RHP zero). $\dot{y} = 0$ for $e^{70t/3} = 13/6$, i.e. $t = 33.1 \text{ ms}$, where $y_{\min} = -0.436$. The exact response is 0 for $t < T_d$ and $1 - e^{-(t-T_d)/T}$ afterwards; the approximation reaches 0.5 already after 173 ms instead of 221 ms (Exercise 7.3).

iv. $\Delta\varphi = \omega T_d - 2 \arctan(\omega T_d/2) \approx (\omega T_d)^3/12$. $\Delta\varphi = 5^\circ = 0.0873$ gives $\omega T_d \approx 1.02$; the exact numerical solution (Exercise 7.3) is $\omega T_d = 1.069$, i.e. $f = 0.85 \text{ Hz}$. Padé(1,1) is fine for controller design at loop frequencies below $\approx 0.5 \text{ Hz}$, useless for the time response around $t = T_d$.

Exercise 7.2 Knock resonator: ODE solvers vs. Laplace

JSLINUX

Program `code/jslinux/sheet07/resonator_ode.c` integrates the ODE $\ddot{y} + 2\sigma\dot{y} + \omega_0^2 y = \omega_0^2 u$ of (1) in state-space form $x = (y, \dot{y})^T$.

- Implement `rhs()`, one step of the forward Euler method and one step of the classical Runge–Kutta method (RK4).
- Implement the analytic impulse and step responses of Exercise 7.1. The program simulates the impulse response (initial state $y(0) = 0$, $\dot{y}(0) = \omega_0^2$ – why?) and the step response for step sizes $h = 40, 20, \dots, 0.625 \mu\text{s}$ and prints the maximum error. Estimate the order of convergence p

from successive errors ($e(h) \propto h^p$).

- c) Explain the results of the Euler method: derive the stability condition $|1 + h\lambda| < 1$ for $\dot{x} = \lambda x$ and evaluate it for the poles of H_K . Compare the limit with the sampling period of the engine chip (20 μs). Why does the chip compute the burst from the closed-form solution instead of integrating an ODE?
- d) Implement `freq_resp()` and compare peak, resonance frequency and -3 dB bandwidth with your results of Exercise 7.1 c).

Solution

a)/b) Key part of `code/solutions/jslinux/sheet07/resonator_ode.c`:

```

1 static void rhs(const double *x, double u, double *dx)
2 {
3     dx[0] = x[1];
4     dx[1] = -w0 * w0 * x[0] - 2.0 * sig * x[1] + w0 * w0 * u;
5 }
6
7 static void euler_step(double *x, double u, double h)
8 {
9     double k[2];
10    rhs(x, u, k);
11    x[0] += h * k[0];
12    x[1] += h * k[1];
13 }
14
15 static void rk4_step(double *x, double u, double h)
16 {
17     double k1[2], k2[2], k3[2], k4[2], xt[2];
18     int i;
19     rhs(x, u, k1);
20     for (i = 0; i < 2; i++) xt[i] = x[i] + 0.5 * h * k1[i];
21     rhs(xt, u, k2);
22     for (i = 0; i < 2; i++) xt[i] = x[i] + 0.5 * h * k2[i];
23     rhs(xt, u, k3);
24     for (i = 0; i < 2; i++) xt[i] = x[i] + h * k3[i];
25     rhs(xt, u, k4);
26     for (i = 0; i < 2; i++)
27         x[i] += h / 6.0 * (k1[i] + 2.0 * k2[i] + 2.0 * k3[i] + k4[i]);
28 }

```

The input $\omega_0^2 \delta(t)$ integrated over $[0^-, 0^+]$ changes $\dot{y}$ by ω_0^2 while y stays continuous, hence the initial state. Program output (maximum error relative to the peak of the exact response, 0–4 ms):

h [μs]	40	20	10	5	2.5	1.25	0.625
Euler	$3.1 \cdot 10^{27}$	$7.8 \cdot 10^{20}$	$2.9 \cdot 10^{11}$	$1.0 \cdot 10^5$	28.7	0.613	0.206
RK4	0.361	$4.6 \cdot 10^{-2}$	$3.0 \cdot 10^{-3}$	$1.8 \cdot 10^{-4}$	$1.1 \cdot 10^{-5}$	$7.2 \cdot 10^{-7}$	$4.5 \cdot 10^{-8}$
RK4 order p	–	2.97	3.96	4.00	4.01	4.00	4.00

RK4 converges with order 4 (error ratio 16 per halving of h) once $\omega_0 h \lesssim 0.4$; at $h = 20\text{ }\mu\text{s}$ (50 samples/ms) the error is 4.6 %. The step response behaves the same way (RK4: $2.3 \cdot 10^{-2}$ at 20 μs , $9.2 \cdot 10^{-5}$ at 5 μs). The Euler method diverges for $h \geq 2.5\text{ }\mu\text{s}$; for $h = 1.25$ and $0.625\text{ }\mu\text{s}$ it is stable but still has 61 % and 21 % error (the observed order 1.57 is still pre-asymptotic, theoretically $p = 1$).

c) Euler applied to $\dot{x} = \lambda x$: $x_{n+1} = (1 + h\lambda)x_n$, stable iff $|1 + h\lambda| < 1$ (a circle of radius 1 around -1 in the $h\lambda$ -plane). With $\lambda = -\sigma \pm j\omega_d$: $|1 + h\lambda|^2 = 1 - 2h\sigma + h^2\omega_0^2 < 1 \iff h < 2\sigma/\omega_0^2 = 1.497\text{ }\mu\text{s}$. For lightly damped poles close to the imaginary axis the forward Euler method is unstable for any reasonable step – the chip samples every 20 μs , more than 13 times this limit. For the poles of H_K , RK4 is stable up to $h\omega_0 \approx 2.8$ ($h \approx 69\text{ }\mu\text{s}$), but still inaccurate. The chip

therefore evaluates the closed-form solution $e^{-t/\tau} \sin(\omega_d t)$ which is exact for every step size – this is precisely what the inverse Laplace transform gives us.

d) Peak 24.27 dB at 6497.0 Hz, -3 dB band 6294.8–6693.1 Hz, $B = 398.3$ Hz ($\sigma/\pi = 397.9$ Hz). Selected values: 0.21 dB at 1 kHz, -4.14 dB at 10.5 kHz, -18.55 dB at 20 kHz, -47.4 dB at 100 kHz (one decade above f_0 about -40 dB plus the small offset of the asymptote); phase -89.1° at 6.5 kHz, -20.8° at 6 kHz, -157.5° at 7 kHz.

Exercise 7.3 Lambda sensor: delay and Padé approximation

JSLINUX

Program `code/jslinux/sheet07/lambda_pade.c` compares the exact step response of $G(s) = e^{-sT_d}/(1 + sT)$ with three rational approximations of the delay: a first-order lag $1/(1 + sT_d)$, Padé(1,1) and Padé(2,2)

$$e^{-sT_d} \approx \frac{1 - sT_d/2 + s^2T_d^2/12}{1 + sT_d/2 + s^2T_d^2/12}.$$

- Any strictly proper $G(s) = N(s)/D(s)$ can be simulated in *controllable canonical form*: $D(p)v = u$ and $y = N(p)v$, where $p = d/dt$ and state $(v, \dot{v}, \dots)$. Implement `ccf_rhs()` and `ccf_out()` (integration by RK4 is given).
- Implement the exact system by a delay line (ring buffer of $D = T_d/h$ samples) followed by the exact discretisation of the lag, $y_{k+1} = a y_k + (1 - a) u_{k-D}$, $a = e^{-h/T}$, and the analytic step response. Enter the Padé coefficients. Compare minimum, time of $y = 0.5$, RMS and maximum error of the three approximations.
- Compare the phase of the delay part. Up to which frequency is the phase error of Padé(1,1) and Padé(2,2) below 5° ?
- Which approximation would you use (i) to design a lambda controller, (ii) to predict the sensor signal in the time domain?

Solution

a)/b) Core of `code/solutions/jslinux/sheet07/lambda_pade.c`:

```
1 static void ccf_rhs(const tf_t *g, const double *x, double u, double *dx)
2 {
3     int i, n = g->n;
4     double s = u;
5     for (i = 0; i < n - 1; i++) dx[i] = x[i + 1];
6     for (i = 0; i < n; i++) s -= g->a[n - i] * x[i];
7     dx[n - 1] = s / g->a[0];
8 }
9
10 static double ccf_out(const tf_t *g, const double *x)
11 {
12     int i, m = g->m;
13     double y = 0.0;
14     for (i = 0; i <= m; i++) y += g->b[m - i] * x[i]; /* b_m x1 + ... */
15     return y;
16 }
```

The delay line reproduces the analytic response to $5.6 \cdot 10^{-15}$. Results (0–1 s, $h = 0.1$ ms):

model	min y	$t_{\min}$ [ms]	t_{50} [ms]	RMS error	max. error
lag $1/(1 + sT_d)$	0.000	–	170.9	0.183	0.567
Padé(1,1)	-0.436	33.1	173.2	0.172	0.616
Padé(2,2)	-0.272	88.8	207.7	0.104	0.448
exact	0	–	220.8	–	–

Padé(1,1) confirms the analytic minimum -0.436 at 33.1 ms of Exercise 7.1. Padé(2,2) (poles

$-15 \pm j8.66$, zeros $+15 \pm j8.66$) has a smaller, oscillating undershoot and a better t_{50} . No rational approximation can produce an output that is exactly zero for $0 < t < T_d$.

c) Phase of the delay part in degrees:

f [Hz]	exact	lag	Padé(1,1)	Padé(2,2)
0.1	-7.2	-7.2	-7.2	-7.2
0.5	-36.0	-32.1	-34.9	-36.0
1.0	-72.0	-51.5	-64.3	-71.8
2.0	-144.0	-68.3	-103.0	-138.7

Error $> 5^\circ$ above 0.85 Hz ($\omega T_d = 1.07$) for Padé(1,1) and above 1.97 Hz ($\omega T_d = 2.48$) for Padé(2,2). The lag has the correct low-frequency phase slope but its phase is bounded by -90° .

d) (i) Controller design (stability margins, Sheet 2 limit cycle) needs the correct *phase* near the crossover frequency; Padé(2,2) (or (1,1) if the crossover is well below 0.8 Hz) is appropriate. (ii) For time-domain prediction one should keep the delay exact (delay line, Smith predictor); every rational approximation produces a non-physical undershoot.

Exercise 7.4 From s to z : discretisation of analog systems

JSLINX

A digital filter shall imitate $H_K(s)$ and $H_{RC}(s)$ at $f_s = 40$ kHz, $T = 1/f_s$. Three classical mappings are

$$\text{forward Euler: } s = \frac{z-1}{T}, \quad \text{bilinear: } s = \frac{2}{T} \frac{z-1}{z+1},$$

$$\text{impulse invariance: } h[n] = T h(nT) \quad (\text{poles } z_k = e^{p_k T}).$$

- (paper)** Into which region of the z -plane is the left half s -plane mapped by each method, and onto which curve the imaginary axis? Which method can turn a stable analog system into an unstable digital one? Show that the bilinear transform maps ω_a onto $\Omega = 2 \arctan(\omega_a T/2)$ (*frequency warping*).
- Complete the three mapping functions of the program `code/jslinux/sheet07/discretise.c`, i.e. `map_euler`, `map_bilin` and `impinv` (second-order section $H(s) = (b_0 s^2 + b_1 s + b_2)/(a_0 s^2 + a_1 s + a_2)$; impulse invariance via partial fractions with the small complex struct provided).
- Prewarping*: with $s = K \frac{z-1}{z+1}$, $K = \omega_r / \tan(\omega_r T/2)$, the frequency ω_r is mapped exactly. Use $\omega_r = \omega_0$ for the resonator and $\omega_r = \omega_c$ for the RC filter.
- Run the program and interpret the tables: pole radius and angle, peak frequency, gain at DC and close to $f_s/2$. Which method would you choose for the knock resonator, which for the anti-aliasing filter model?

Solution

a) Forward Euler: $z = 1 + sT$; the left half plane becomes the half plane $\text{Re } z < 1$, the $j\omega$ axis the vertical line $\text{Re } z = 1$. Stable poles outside the unit circle become unstable (exactly the stability condition of Exercise 7.2). Impulse invariance: $z = e^{sT}$, LHP $\rightarrow$ interior of the unit circle (stable stays stable), but the map is not one-to-one: all $s + jk2\pi/T$ map to the same $z \Rightarrow$ aliasing of the frequency response. Bilinear: $z = (1 + sT/2)/(1 - sT/2)$ maps the LHP one-to-one onto the unit disk and the $j\omega$ axis onto the unit circle; with $z = e^{j\Omega}$: $\frac{2}{T} \frac{e^{j\Omega}-1}{e^{j\Omega}+1} = j \frac{2}{T} \tan \frac{\Omega}{2} = j\omega_a$, hence $\Omega = 2 \arctan(\omega_a T/2)$; the whole axis $0 \leq \omega_a < \infty$ is compressed into $0 \leq \Omega < \pi$.

b) Key part of the solution:

```
1 static void map_euler(const double *c, double T, double *C)
2 {
3     C[0] = c[0] / (T * T);
4     C[1] = -2.0 * c[0] / (T * T) + c[1] / T;
5     C[2] = c[0] / (T * T) - c[1] / T + c[2];
```

```
6 }
```

```
1 static void map_bilin(const double *c, double K, double *C)
2 {
3     C[0] = c[0] * K * K + c[1] * K + c[2];
4     C[1] = -2.0 * c[0] * K * K + 2.0 * c[2];
5     C[2] = c[0] * K * K - c[1] * K + c[2];
6 }
```

```
1 static dig_t impinv(const ana_t *h, double T)
2 {
3     dig_t d;
4     if (h->a[0] == 0.0) { /* first order b2 / (a1 s + a2) */
5         double p = -h->a[2] / h->a[1], r = h->b[2] / h->a[1];
6         d.B[0] = T * r; d.B[1] = 0; d.B[2] = 0;
7         d.A[0] = 1.0; d.A[1] = -exp(p * T); d.A[2] = 0;
8     } else { /* second order, distinct poles */
9         cpx disc = cx(h->a[1] * h->a[1] - 4 * h->a[0] * h->a[2], 0);
10        cpx sq = csqrt(disc);
11        cpx p1 = cscale(cadd(cx(-h->a[1], 0), sq), 0.5 / h->a[0]);
12        cpx p2 = cscale(csub(cx(-h->a[1], 0), sq), 0.5 / h->a[0]);
13        /* residue r_i = N(p_i) / D'(p_i), D'(s) = 2 a0 s + a1 */
14        cpx N1 = cadd(cadd(cscale(cmul(p1, p1), h->b[0]), cscale(p1, h->b[1])), cx(h->b[2],
15        0));
16        cpx N2 = cadd(cadd(cscale(cmul(p2, p2), h->b[0]), cscale(p2, h->b[1])), cx(h->b[2],
17        0));
18        cpx r1 = cdiv(N1, cadd(cscale(p1, 2 * h->a[0]), cx(h->a[1], 0)));
19        cpx r2 = cdiv(N2, cadd(cscale(p2, 2 * h->a[0]), cx(h->a[1], 0)));
20        cpx q1 = cexpo(cscale(p1, T)); q2 = cexpo(cscale(p2, T));
21        /* T [ r1 (1 - q2 z^-1) + r2 (1 - q1 z^-1) ] / [(1 - q1 z^-1)(1 - q2 z^-1)] */
22        d.B[0] = T * cadd(r1, r2).re;
23        d.B[1] = -T * cadd(cmul(r1, q2), cmul(r2, q1)).re;
24        d.B[2] = 0.0;
25        d.A[0] = 1.0;
26        d.A[1] = -cadd(q1, q2).re;
27        d.A[2] = cmul(q1, q2).re;
28    }
29    return d;
30 }
```

c)/d) Program output, knock resonator (exact pole map: $|z| = e^{-\sigma T} = 0.96923$ at 6500 Hz):

method	$ p $	$\arg p$ [Hz]	peak [Hz]	$ H $ at 0	at 6.5 kHz	at 15 kHz
analog	—	—	6497	0.00 dB	24.27 dB	-12.72 dB
forward Euler	1.4075	5167	(4801)	0.00	1.11	-11.90
impulse invariance	0.9692	6500	6498	-0.80	24.27	-9.02
bilinear	0.9755	6011	6008	0.00	13.43	-26.59
bilinear, prewarped	0.9742	6502	6498	0.00	24.27	-24.89

Coefficients: impulse invariance $H(z) = 0.844567 z^{-1} / (1 - 1.012846 z^{-1} + 0.939413 z^{-2})$, bilinear $H(z) = 0.201889(1 + z^{-1})^2 / (1 - 1.144074 z^{-1} + 0.951630 z^{-2})$. Forward Euler produces poles at $|z| = 1.41$: **unstable** (its “frequency response” is a formal evaluation without meaning). Impulse invariance keeps the resonance perfectly (pole angle exactly 6500 Hz, peak 24.27 dB) but aliasing raises the response far from the peak (-0.8 dB at DC, -9.0 dB instead of -12.7 dB at 15 kHz). The plain bilinear transform warps the resonance to $f_s / \pi \cdot \arctan(\pi f_0 / f_s) = 6.01$ kHz - 500 Hz off, more than the bandwidth, so at 6.5 kHz only 13.4 dB instead of 24.3 dB remain. Prewarping at f_0 moves the pole back to 6502 Hz. The bilinear methods have a double zero at $z = -1$ ($|H| \rightarrow -\infty$ at $f_s/2$).

RC low-pass ($f_c = 15.9$ kHz, $\omega_c T = 2.5$):

method	pole z	$ H $ at 0	6.5 kHz	15.9 kHz	20 kHz
analog	–	0.00 dB	–0.67 dB	–3.01 dB	–4.11 dB
forward Euler	–1.50 (unstable)	0.00	1.13	8.68	13.98
impulse invariance	0.0821	8.70	8.32	7.40	7.27
bilinear	–0.111	0.00	–0.79	–8.32	–80
bilinear, prewarped	–0.501	0.00	–0.15	–3.01	–73

For a filter whose response does not vanish below $f_s/2$, impulse invariance is useless: the DC gain is $\omega_c T / (1 - e^{-\omega_c T}) = 2.72$ (8.7 dB), the sum of all aliased copies. Choice: for the knock resonator (narrow-band, $f_0 \ll f_s/2$) impulse invariance or prewarped bilinear; for the RC model (and any low-/high-pass or band-stop) the prewarped bilinear transform. This is the standard IIR design route of Sheet 10.

Deliverables

Paper solution of 7.1 including the Bode sketch; the error table and the order estimate of 7.2 with the Euler stability limit; the step-response table of 7.3 and your recommendation 7.3 d); the comparison tables of 7.4 with a short justification of your choice of discretisation method.